

ASTRAKHANKA, Russian Federation

WMO: 319210

Lat: 44.7169N

Lon: 132.0730E

Elev: 78

StdP: 100.39

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-23.5	-30.1	0.2	-24.4	-28.0	0.3	-22.0	10.0	-10.8	8.0	-11.3	1.5	220	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	29.0	23.1	27.4	22.1	26.1	21.4	24.3	27.5	23.3	26.2	22.4	25.0	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.2	26.3	22.3	17.2	25.3	21.5	16.3	24.3	74.1	27.7	70.2	26.3	66.5	25.0	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.7	6.6	-28.9	32.4	3.4	1.3	-31.3	33.3	-33.3	34.0	-35.3	34.8	-37.7	35.7		
(5)			-29.0	26.1	3.4	1.9	-31.5	27.5	-33.4	28.6	-35.3	29.7	-37.8	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-15.9	-11.3	-3.2	5.8	12.8	17.8	21.5	21.2	15.5	7.4	-3.2	-13.0		
	DBStd	13.69	4.62	5.02	5.06	4.06	3.39	2.83	2.40	2.49	3.38	4.38	5.99	4.76		
	HDD10.0	3175	804	597	410	138	11	0	0	0	2	103	397	712		
	HDD18.3	5212	1063	831	668	376	173	42	3	6	96	338	646	971		
	CDD10.0	1239	0	0	0	12	99	236	355	346	167	24	1	0		
	CDD18.3	234	0	0	0	0	3	27	101	93	11	0	0	0		
	CDH23.3	1216	0	0	0	4	46	171	534	413	45	1	0	0		
	CDH26.7	218	0	0	0	0	9	31	113	61	4	0	0	0		
(14) Wind	WSAvg	3.2	2.4	2.9	3.4	3.8	3.9	3.6	3.2	3.0	3.0	3.3	3.2	2.7		
(15) Precipitation	PrecAvg	581	9	10	19	35	70	72	106	111	73	49	21	14		
	PrecMax	765	36	28	57	77	134	139	217	214	181	127	43	35		
	PrecMin	383	1	1	1	7	21	15	20	10	11	14	5	3		
	PrecStd	107	9	8	13	21	33	30	49	50	46	27	13	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	4.5	13.4	23.3	27.8	29.9	31.3	30.5	26.8	21.5	14.1	3.5		
		MCWB	-2.4	1.3	6.7	13.0	17.6	21.9	24.9	24.0	20.3	14.5	8.6	1.7		
	2%	DB	-4.1	1.5	8.2	18.4	24.1	27.0	29.4	28.2	24.5	18.5	10.4	-0.4		
		MCWB	-5.6	-0.4	3.6	10.3	15.5	20.7	24.0	22.9	19.2	13.2	6.8	-1.8		
	5%	DB	-6.7	-0.6	5.2	15.0	21.4	25.2	27.8	26.9	22.8	16.1	7.5	-3.3		
		MCWB	-8.1	-2.5	1.9	8.9	14.5	19.9	23.0	22.5	17.9	11.9	4.7	-4.7		
	10%	DB	-8.9	-3.3	3.3	12.3	19.1	23.5	26.2	25.6	21.1	14.3	5.0	-5.7		
		MCWB	-10.1	-5.0	0.8	7.6	13.7	18.9	22.1	21.7	17.2	11.1	2.5	-7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.0	1.8	7.1	13.8	18.9	23.5	26.5	25.3	22.1	16.1	10.0	2.1		
		MCDB	-1.2	3.5	12.7	21.1	26.0	27.5	29.7	28.7	24.7	19.3	12.6	3.0		
	2%	WB	-5.6	-0.2	4.1	11.3	16.7	21.7	24.7	24.2	20.3	14.4	7.2	-1.8		
		MCDB	-4.1	1.3	7.5	16.9	22.2	25.5	28.2	27.1	23.1	17.0	10.0	-0.6		
	5%	WB	-8.0	-2.5	2.4	9.7	15.3	20.4	23.6	23.4	19.1	13.1	5.0	-4.7		
		MCDB	-6.7	-0.7	4.6	14.2	20.1	24.3	26.8	25.8	21.6	15.5	7.1	-3.5		
	10%	WB	-10.1	-5.0	1.2	8.1	14.1	19.3	22.6	22.5	18.0	11.4	2.7	-7.1		
		MCDB	-8.9	-3.3	3.1	12.0	18.1	23.0	25.5	24.7	20.4	13.7	4.7	-5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.8	8.4	8.6	9.2	8.8	7.6	7.3	9.2	9.7	8.6	8.5		
		MCDBR	11.2	11.5	10.8	14.5	14.6	12.3	10.5	9.1	11.2	12.6	12.2	10.8		
		MCWBR	9.9	9.8	7.4	8.2	7.5	6.8	5.7	4.9	6.3	7.9	9.0	9.5		
		MCDBR	11.2	11.1	10.2	13.3	12.4	11.0	9.4	7.9	8.9	10.5	11.5	10.6		
	5% WB	MCWBR	10.0	9.5	7.3	8.0	7.2	6.7	5.7	4.8	5.6	7.2	9.1	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.371	0.456	0.567	0.582	0.563	0.522	0.482	0.437	0.459	0.389	0.336		
	taud		2.057	1.920	1.829	1.694	1.730	1.863	2.026	2.133	2.158	1.952	2.006	2.021		
	Ebn at Noon		753	774	755	706	714	731	757	775	775	672	656	680		
	Edh at Noon		103	144	182	226	225	198	166	144	130	139	109	97		
(44) All-Sky Solar Radiation	RadAvg		2.18	3.27	4.36	4.63	5.19	5.57	4.99	4.69	4.20	2.99	2.04	1.75		
	RadStd		0.14	0.15	0.33	0.42	0.43	0.59	0.49	0.40	0.34	0.23	0.16	0.09		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page